GOTYME BANKING SYSTEM

A TEST CASE FOR FUNCTIONALITY, EXPLORATORY, AND SECURITY FOR QUALITY ASSURANCE

A Final Requirement Presented to the

Faculty of the College of Computing and Multimedia Studies

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Camarines Norte State College

in Partial Fulfillment of the Requirements

for the Course of the System Quality Assurance

In program of Information Systems

by

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MAY 2025

System Software Name:

GoTyme Bank System

System's Purpose:

GoTyme Bank is a digital banking platform in the Philippines that aims to make banking easy, safe, and accessible for everyone. It allows users to create accounts, save money, send and receive funds, and manage their finances through a mobile app. GoTyme also offers physical kiosks and debit cards for added convenience. With a focus on both technology and personal support, the bank provides a smooth digital experience while still offering customer assistance when needed.

Test Environment:

Devices: Android smartphone

Network: 4G LTE and Wi-Fi

Testing Tools: Manual Testing

Users: Testing a user account

System's Objective

To provide fast, secure, and easy-to-use digital banking services that allow users to manage their money anytime, anywhere. The system aims to improve financial access and customer satisfaction through digital innovation.

FUNCTIONAL TESTING

Test Scenario ID	Test Case Description
TS_001	As a user, I want to create a Go Tyme account so I can start digital banking
TS_002	User enters correct passcode to access their GoTyme account
TS_003	As a User, I want to change my passcode so I can only access my account.
TS-004	As a user, I want to view my account balance so I can track my money.
TS_005	As a user, I want to send money to other banks so I can transfer funds easily.
TS_006	As a User, I want to send money to a friend's GoTyme account.

TS_007	As a user, I want to access my transaction history to verity transaction
TS_008	As a user, I want to log out of my account so I can protect my privacy.
TS_009	As a user, I want to set up fingerprint login so I can log in faster.
TS_010	As a user, I want to create my “Go Save” account. So that I can view and manage my savings effectively.
TS_011	As a user, I want to scan a QR code, So that I can make payments or send money easily without manually inputting details.
TS_012	As a user, I want to buy prepaid mobile load, So that I can easily top-up my phone or someone else’s.
TS_013	As a user, I want to set up a USD time deposit, So that I can save money in USD and earn interest over time.
TS-014	As a User, I want to change my registered email address from the settings
TS_015	As a user, I want to check if the system still go if there’s internet interruptions.

CaseID	Description	Expected Outcome	Actual Outcome	Remarks
Case_001	Verify Account Creation Functionality	User Created an Account	Directed to Homepage	Passed
Case_002	Verify login Functionality	User successfully login and Directed to the Homepage	Directed to the Homepage	passed
Case_003	Verify Forget Passcode	Resets Passcode	Passcode Reset	Passed
Case_004	View account Balance	Account balance is displayed accurately	See the current Balance	Passed
Case_005	Verify Money transfer	Funds are transferred and user receives confirmation	successful money transfer	Passed
Case_006	Verify Receive Money	Successfully Received Money	Successfully Received Money	passed
Case_007	Verity transaction History	All transaction are recorded	All transaction are recorded	Passed
Case_008	Verify logout functionality	logout Account	successful logout	Passed
Case_009	Verify Biometric functionality	Enable Biometrics	Succesful Register Biometrics	passed

Case_010	Verify Savings functionality	Successful Creation of saving account	Goal is created and appears in the ‘Go Save’ section	Passed
Case_011	Verity Scanner functionality	successful QR Scanning	successfully scanned QR	Passed
Case_012	Verity Buy load Functionality	Buy load Successfully	load successfully and confirmed	Passed
Case_013	Verity USD Time Deposit	Successfully creates time deposit account	time deposit is created	Passed
Case_014	Verify update information	System saves updated information	Successfully Updated information	Passed
Case_015	Verify Points Credit	Points are credited	Points are used and deducted	Passed

Observation

All test cases passed, and the majority of these cases represent the most commonly used features by users. Nonetheless, the verification of the application, GoTyme, demonstrates excellent behavior for a banking system. From the creation of an account, it incorporates a strong choice of security. It employs not only one security measure, such as OTP, but also includes facial recognition, and the app does not permit any screenshots during the facial recognition process. It acts swiftly; for instance, when using the scanner, I almost not handle a documentation as it takes only milliseconds to scan the QR code. It is noteworthy that I scanned it on a glare-prone screen. Most of its functionalities work well and are impressive. Additionally, all logs and transactions are recorded in real-time within the transaction history, and the system notifies the user at the conclusion of each transaction. In summary, GoTyme possesses robust functionalities and operates effectively even with multiple taps, clearly recording every log; thus, it qualifies as a successful eBank system mobile application.

EXPLORATORY TESTING

Test Scenario ID	**Test Case Description**
TS_001	User opens the app while on airplane mode
TS_002	User quickly taps through multiple sections to test stability
TS_003	User tries to pay bills with below minimum
TS_004	User sets phone to dark mode and opens the app

TS_005	User scrolls through 3 months of transaction logs
TS_006	User logs in from another mobile while already logged in
TS_007	Users tries to input invalid credentials
TS_008	User tries to send money exceeding the account balance
TS-009	User start processing transaction and back mid-process
TS_010	User leave the application and use another application

CaseID	**Description**	**Expected Outcome**	**Actual Outcome**	**Remarks**
Case_001	Navigate app with airplane mode enabled	App shows offline status and disables transactions	App shows lost connection	Passed
Case_002	Rapidly tap everything in the dashboard	App remains stable and responsive, no crash or data loss	App response on everytap without crashing	Passed
Case_003	Pay below the required value in the Pay Bills	App will display warning and the required amount	App display red notice and show required minimum amount	Passed
Case_004	Use app in dark mode system setting	App adjusts interface to dark mode or maintains readability	App do not adjust on dark mode	Failed
Case_005	Scroll quickly through a long transaction history	App loads data seamlessly without freezing	App scroll flowless and don’t seam to be in trouble	Passed
Case_006	Log in with active session on another device	App shows appropriate message or logs out previous session	the other device is cracking, hence, it don’t show real results	Not Respondin g
Case_007	Login with invalid credentials	App show contimation of invalid input	App do not proceeds and highlighted in red	Passed
Case_008	Send money that is higher than the account balance	App display warning or not enough balance	App displayed warning and notice	Passed
Case_009	Tap ‘Back’ during an ongoing transaction	App display confirmation and exit the transaction	App show confirmation of exiting transaction	Passed
Case_010	Exit Gotyme without logging out then use another app.	App will automatically close	App still open	Failed

Observation

With the exploration of Gotyme, it was discovered that some of its functionalities failed during this assessment. Starting with the dark mode of the system, it does not follow or adjust to what the system conveys, making it less effective in terms of usability regarding visual accessibility. Although it has attractive functionality, it still needs to be more user-centered for broader general usage. In terms of the other explored features and functionalities, it demonstrates great smoothness, and the error handling is exceptional. One of the questions now is the validation of the double account, as documented; the application did not respond when taking the steps to log in, which is why it was noted as not responding. The app also proceeds and maintains the continuation boxes to ensure actions are undertaken. In case number 10, the app did not close after using another application. In some of our testing, it does lock and requires re-login; the assumption is that Gotyme operates under a time constraint before locking the application. Overall, it still meets the exploratory quota for the test.

SECURITY TESTING (MANUAL TESTING)

Test Case ID	**Test Case Description**
TS_001	User intentionally enters wrong password 5 times
TS_002	User logs in, leaves app idle, and checks auto-logout
TS_003	User logs in, switches apps, and returns via recent app
TS_004	User disables fingerprint in phone settings after setting it in-app
TS_005	User records screen during PIN entry or transaction
TS_006	User copies account number and checks if it remains accessible
TS_007	User enters a previously received OTP to try to authenticate
TS_008	User tries to take a screenshot during login, PIN entry, or transaction

CaseID	**Description**	**Expected Outcome**	**Actual Outcome**	**Remarks**
Case_001	Attempt login with incorrect password multiple times	Account is temporarily locked	The account has been locked and need to reset passcode	Passed
Case_002	Leave app idle after login and return after 5 minutes	App requires re-authentication or displays timeout	App closed before even reading four minutes and need to login again	Passed
Case_003	Bypass login via recent apps or task switcher	App should not show anything and required to login again	App do not show information however it needs under for minutes use of other	Failed

			application before requiring to login again	
Case-004	Enable biometric login, then disable it in device settings	App should prompt fallback authentication	App need to re-register the biometries	Passed
Case_005	Use screen recording during sensitive activities	App should warn or block recording	App do not hide sensitive information it caught in the recording	Failed
Case_006	Check clipboard behavior after copying account number	Clipboard should clear sensitive data shortly after copy	Clipboard do not automatically erase the data after being used	Failed
Case_007	Attempt to bypass OTP by reusing an old code	OTP won't work as it was only for single use	OTP do not work	Passed
Case_008	Check if app prevents screenshot during sensitive activities	Warns about screenshot attempts	It only prevents on biometrics but not on other information	Failed

Observation

The security test is the most time-consuming, as our assessments require multiple actions and prerequisites before observing the actual outcome. The only observation regarding security pertains to privacy protection. Although the phone system can block the view when switching between recently used applications, the application itself exhibits some deficiencies in privacy protection. During screen recording, the only portion that turns pitch black is the biometric input; however, when entering the passcode, it remains completely visible. Furthermore, taking screenshots is possible, with the only privacy measure observed being during facial recognition. While security is adequate, there are no security measures that are entirely foolproof.

Recommendations

One of the recommendations throughout the test analysis is symmetric encryption. It is very convenient to have no concerns about waiting for an OTP to conduct bank transfers or send money. It is favorable for those who do not have a strong receipt signature in their area; however, this may present a flaw, as if others gain knowledge of your passcode, they might exploit that information to transfer money without your consent. Nonetheless, it remains a strong point and an area for improvement. Security needs to be more firmly established. More likely, the recommendation pertains to privacy, as it has been observed that the privacy measures of Go Tyme are lacking. Implementing robust data privacy measures is highly recommended to secure banking

usage, as privacy protection is paramount. For instance, a shorter time span to re-enter the passcode when switching applications is advisable. Only fifty percent of the security tests passed, indicating that security requires thorough evaluation.

Conclusion

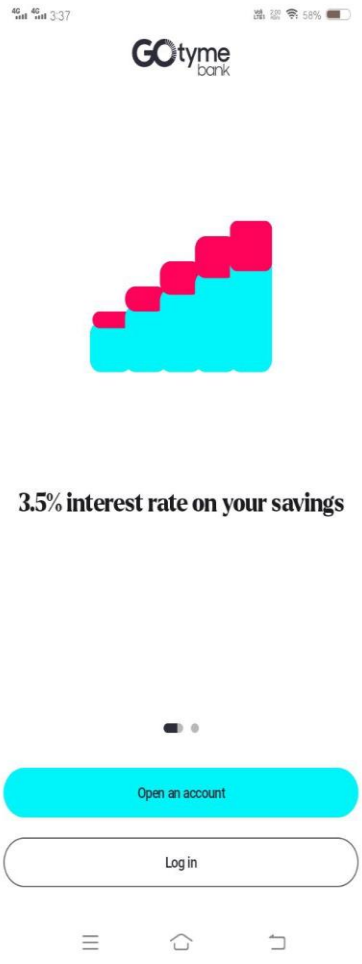
The GoTyme Bank is truly worth considering, offering zero convenience fees for purchasing load for telecommunications, with only an eight-peso charge for bank transfers; subsequent transfers are entirely free. Savings accounts feature an annual interest rate of 3.5%, which is compounded at the end of each month since the initial deposit. Additionally, USD time deposits are available, and surprisingly, points can be redeemed for actual currency. GoTyme provides numerous features and conveniences in banking, including free card printing and opportunities to earn more points. However, GoTyme does have some shortcomings regarding data privacy and disclosure. Security of information is crucial; nonetheless, this issue does not significantly undermine the overall functionality of the application. It remains a solid option with considerable potential for further development.

DOCUMENTATION:



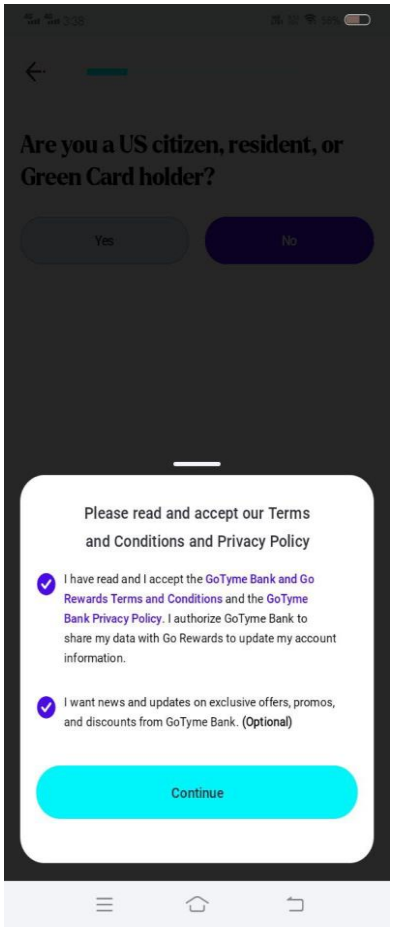
FUNCTIONAL TESTING

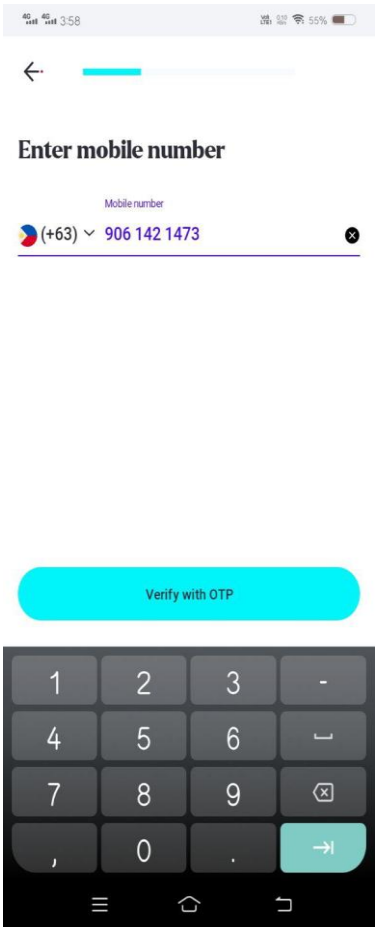
CREATE ACCOUNT

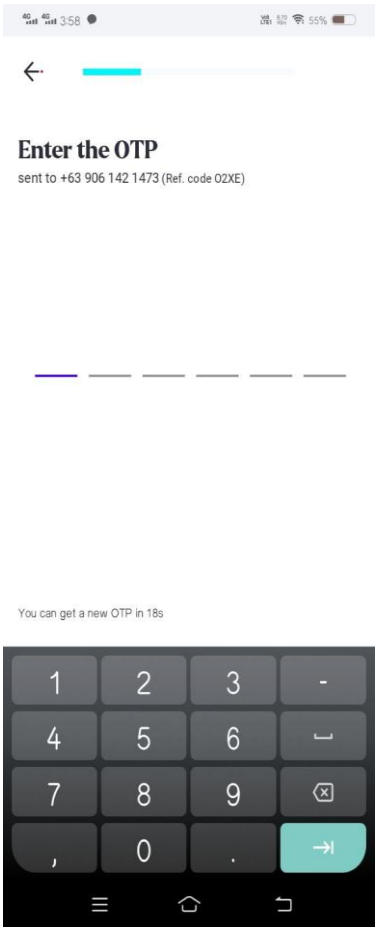


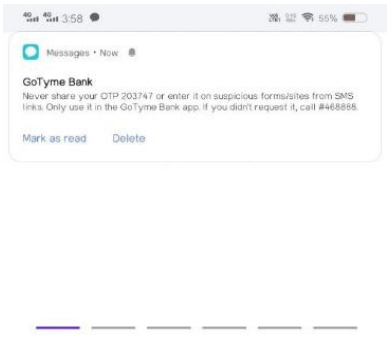


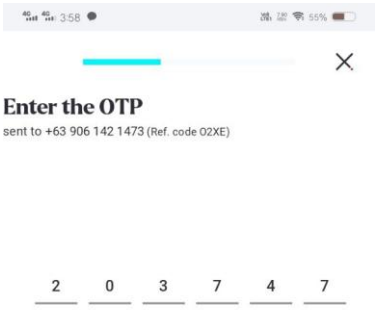


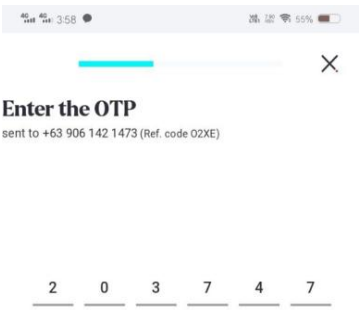

Are you a US citizen, resident, or Green Card holder?
Yes
No
Please read and accept our Terms and Conditions and Privacy Policy
I have read and I accept the GoTyme Bank and Go Rewards Terms and Conditions and the GoTyme Bank Privacy Policy. I authorize GoTyme Bank to share my data with Go Rewards to update my account information.
I want news and updates on exclusive offers, promos, and discounts from GoTyme Bank. (Optional)
Continue


Enter mobile number
Mobile number
(+63)
906 142 1473
Verify with OTP


Enter the OTP
sent to +63 906 142 1473 (Ref. code 02XE)
You can get a new OTP in 18s


Messages • Now
GoTyme Bank
Never share your OTP 203747 or enter it on suspicious forms/sites from SMS links. Only use it in the GoTyme Bank app. If you didn't request it, call #468888.
Mark as read
Delete

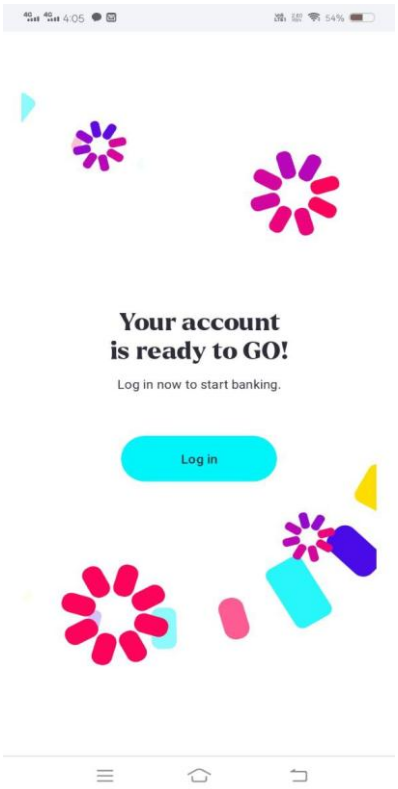

Enter the OTP
sent to +63 906 142 1473 (Ref. code 02XE)
2 0 3 7 4 7

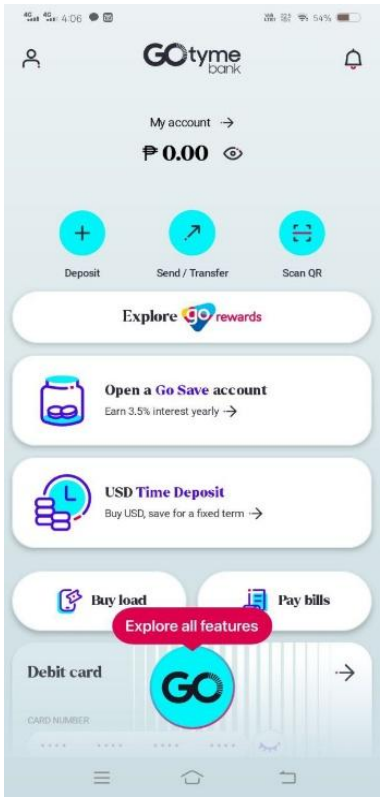

Enter the OTP
sent to +63 906 142 1473 (Ref. code 02XE)
2 0 3 7 4 7

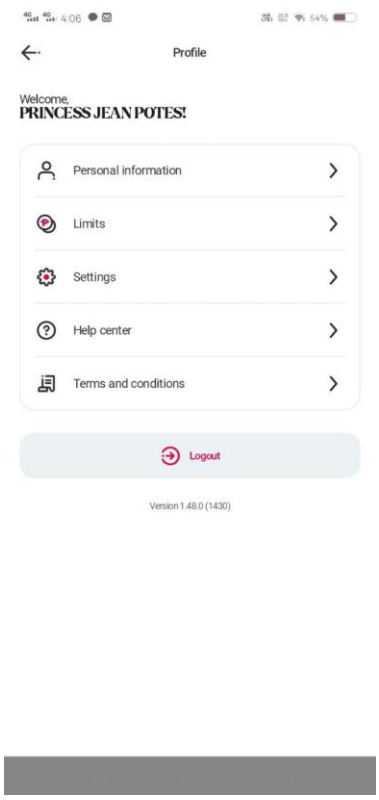

You can get a new OTP in 24s

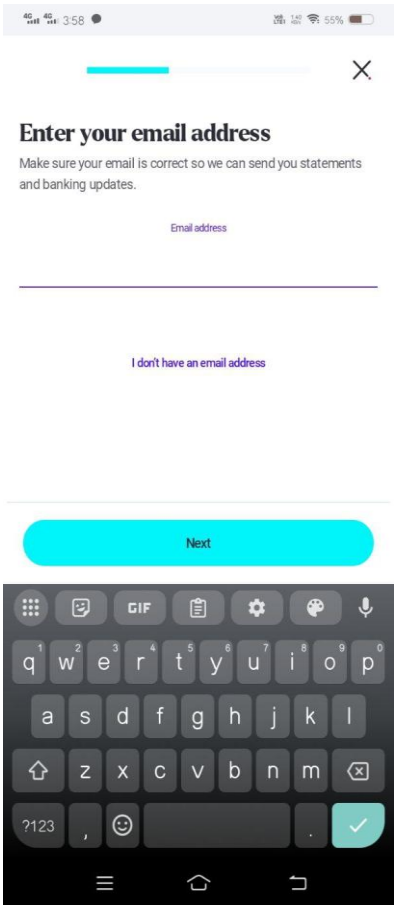

You can get a new OTP in 7s


You can get a new OTP in 7s

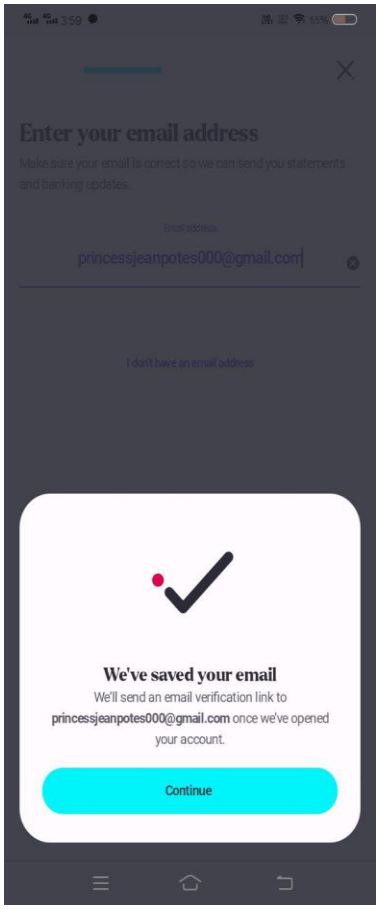

Your account
is ready to GO!
Log in now to start banking.
Log in

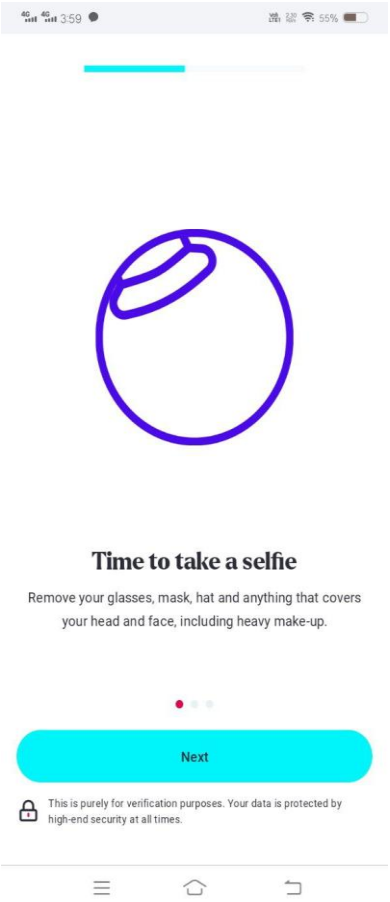

GOtyme bank
My account
₱ 0.00
Deposit
Send / Transfer
Scan QR
Explore rewards
Open a Go Save account
Earn 3.5% interest yearly
USD Time Deposit
Buy USD, save for a fixed term
Buy load
Pay bills
Explore all features
Debit card
CARD NUMBER

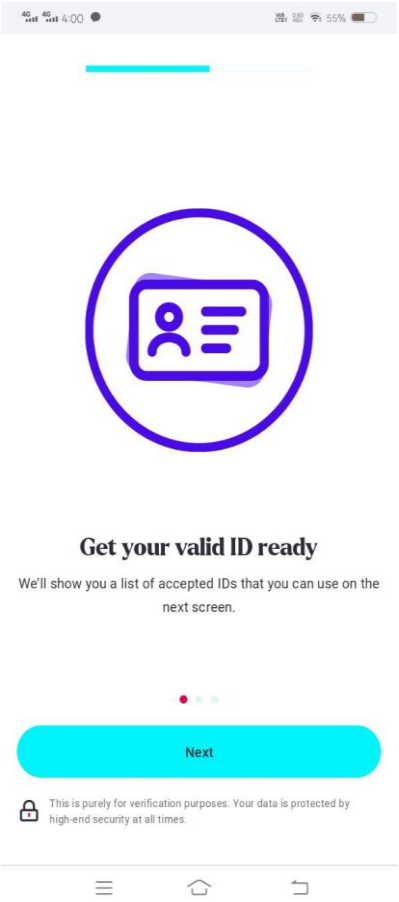

Profile
Welcome,
PRINCESS JEAN POTES!
Personal information
Limits
Settings
Help center
Terms and conditions
Logout
Version 1.48.0 (1430)

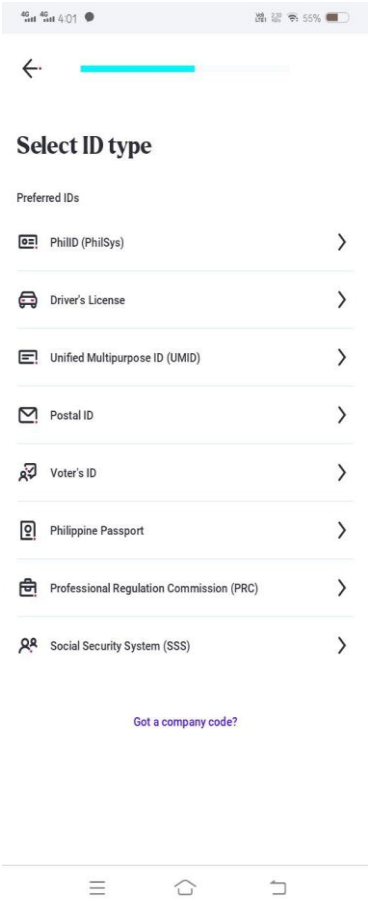

Enter your email address
Make sure your email is correct so we can send you statements and banking updates.
Email address
I don't have an email address
Next


Enter your email address
Make sure your email is correct so we can send you statements and banking updates.
Email address
princessjeanpotes000@gmail.com
I don't have an email address
Next


Enter your email address
princessjeanpotes000@gmail.com
We've saved your email
We'll send an email verification link to princessjeanpotes000@gmail.com once we've opened your account.
Continue

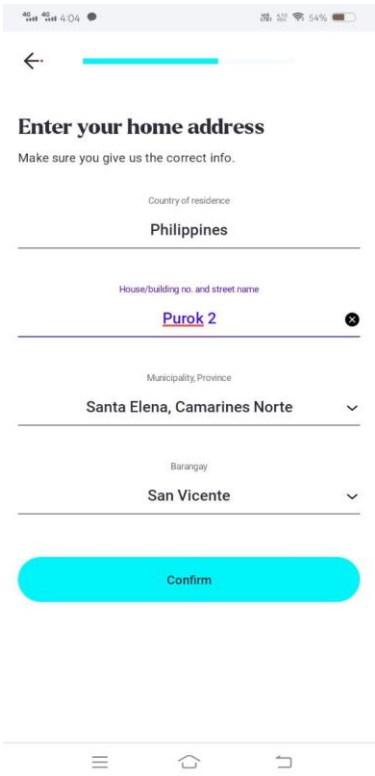

Time to take a selfie
Remove your glasses, mask, hat and anything that covers your head and face, including heavy make-up.
Next
This is purely for verification purposes. Your data is protected by high-end security at all times.


Get your valid ID ready
We'll show you a list of accepted IDs that you can use on the next screen.
Next
This is purely for verification purposes. Your data is protected by high-end security at all times.

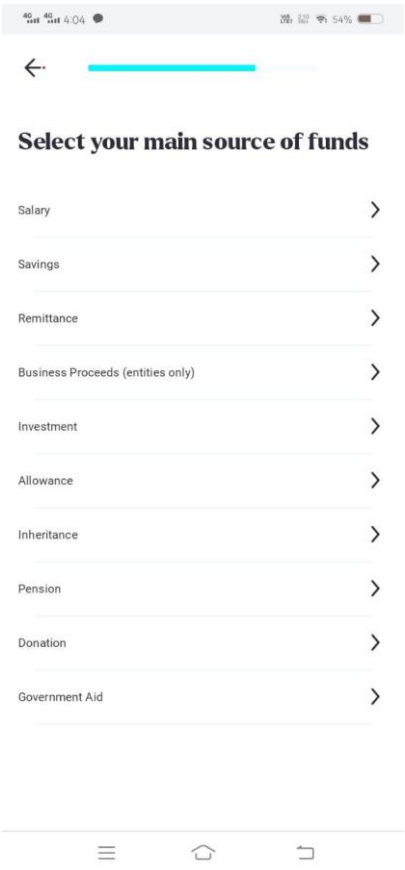

Select ID type
Preferred IDs
PhilID (PhilSys)
Driver's License
Unified Multipurpose ID (UMID)
Postal ID
Voter's ID
Philippine Passport
Professional Regulation Commission (PRC)
Social Security System (SSS)
Got a company code?

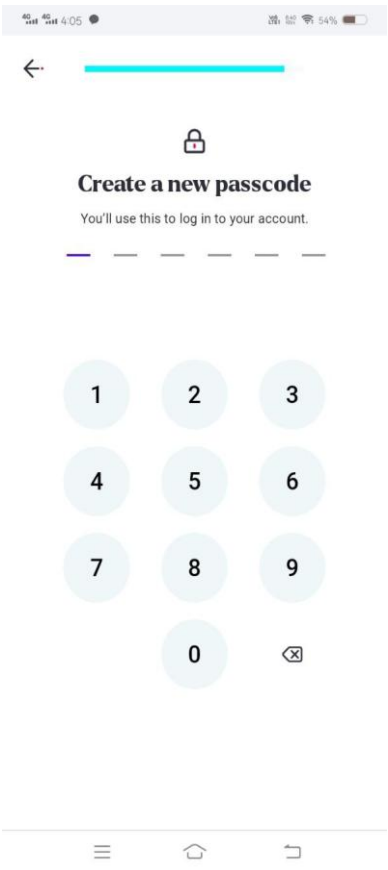

Photos uploaded!
Just a few steps to go


Are these details correct?
ID number
3073-2901-4817-6856
Full name
PRINCESS JEAN PORLAY POTES
No need to re-scan if the following were removed:
ñ, hyphen (-), dot (.), apostrophe (')
Date of birth
November 09 2003
Yes, that's correct
No, re-scan my ID

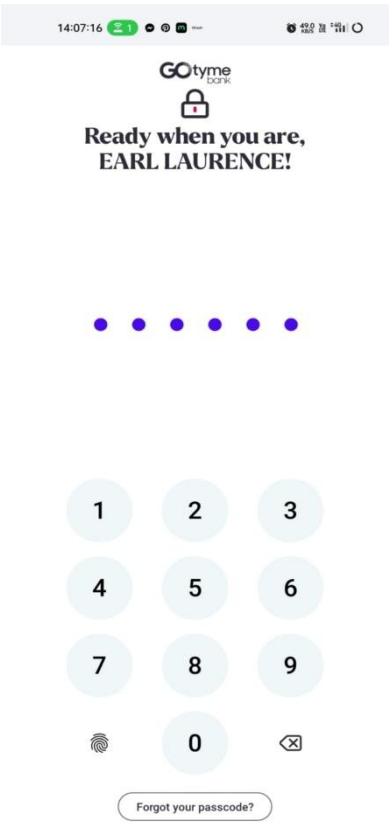

Enter your home address
Make sure you give us the correct info.
Country of residence
Philippines
House/building no. and street name
Purok 2
Municipality, Province
Santa Elena, Camarines Norte
Barangay
San Vicente
Confirm

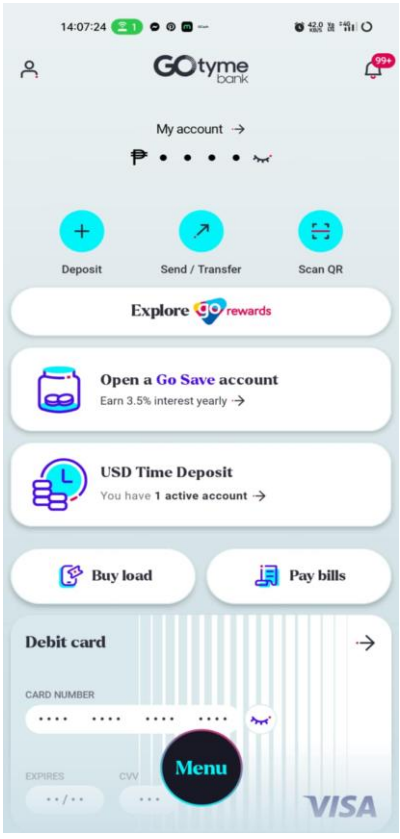




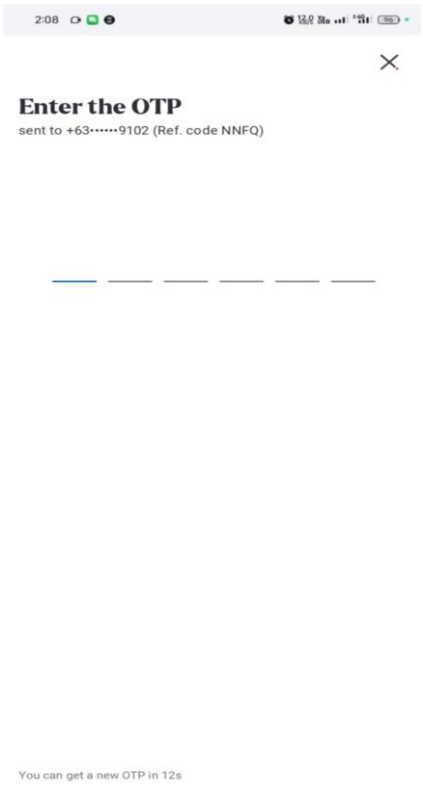


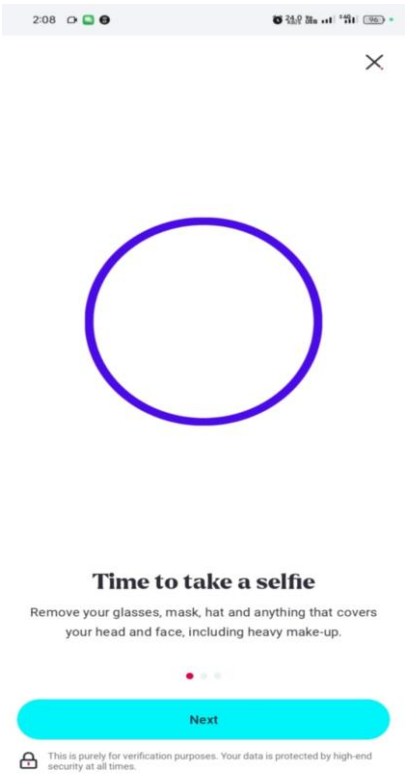
LOGIN FUNCTIONALITY

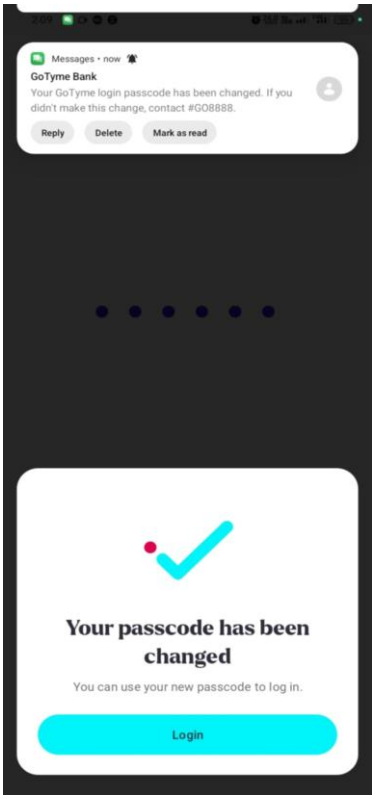




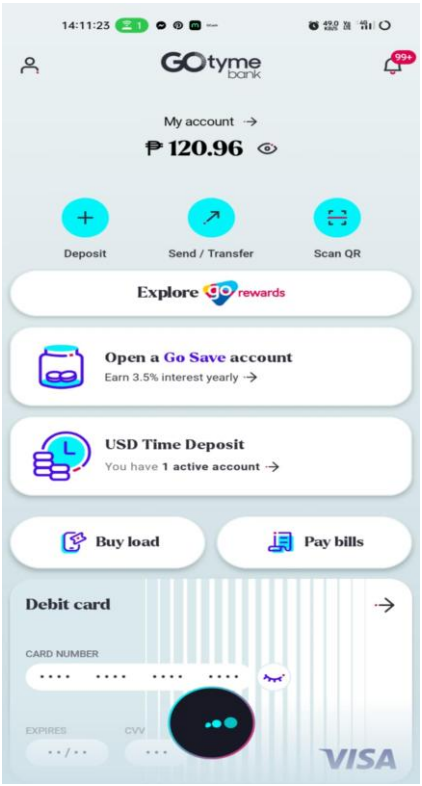
FORGOT PASSCODE



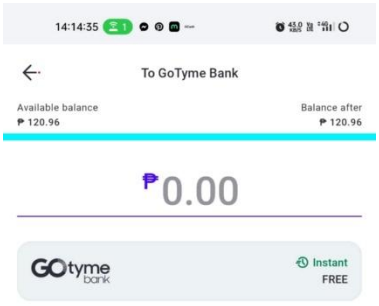


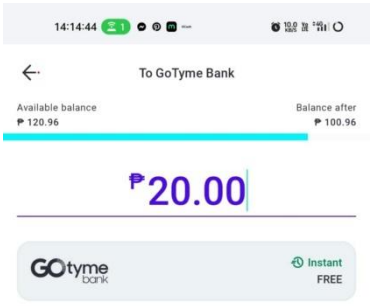


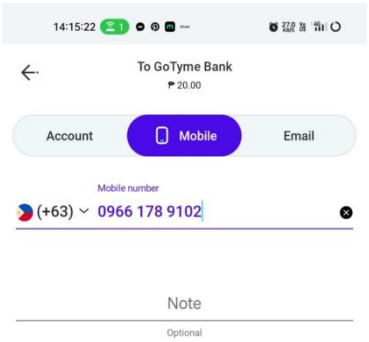
BANK ACCOUNT BALANCE

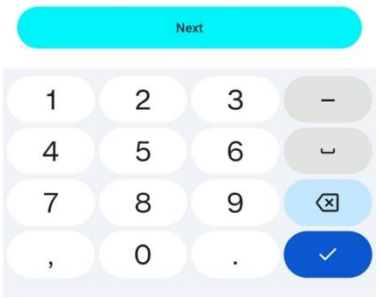


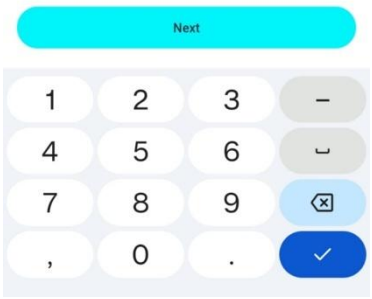
MONEY TRANSFER

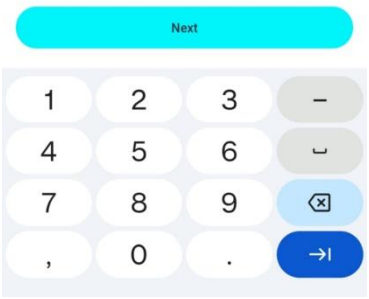


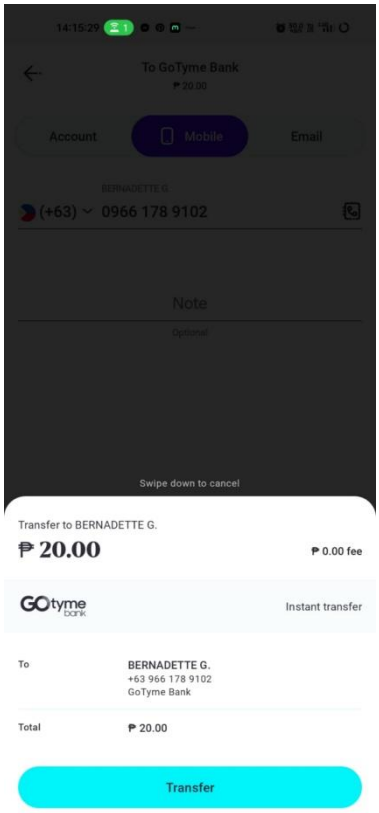


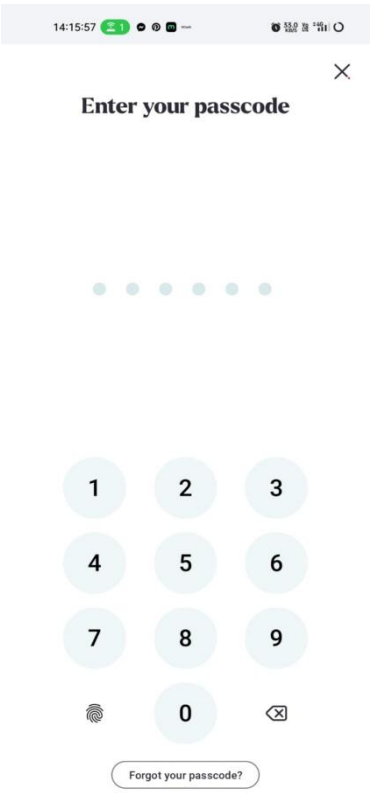






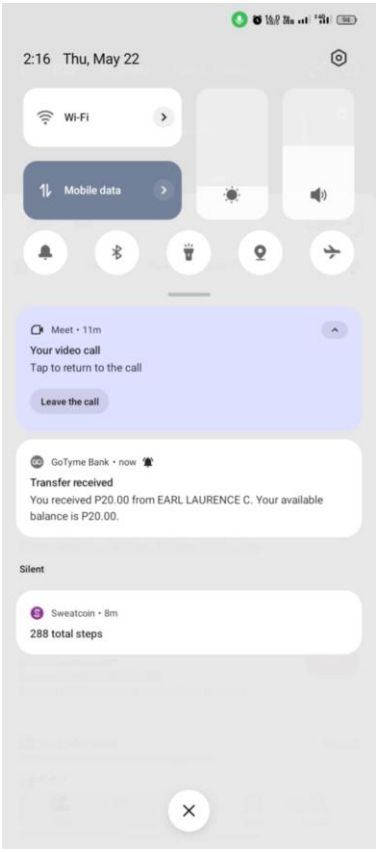


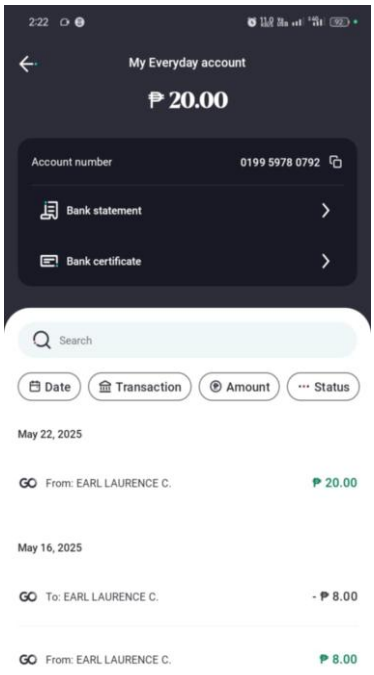




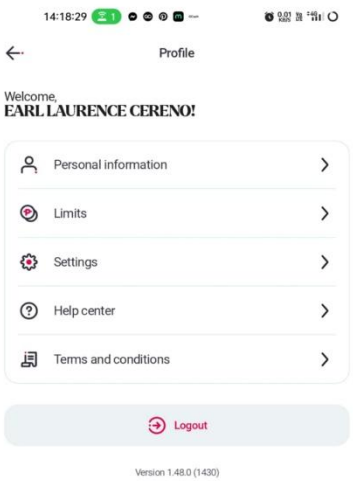


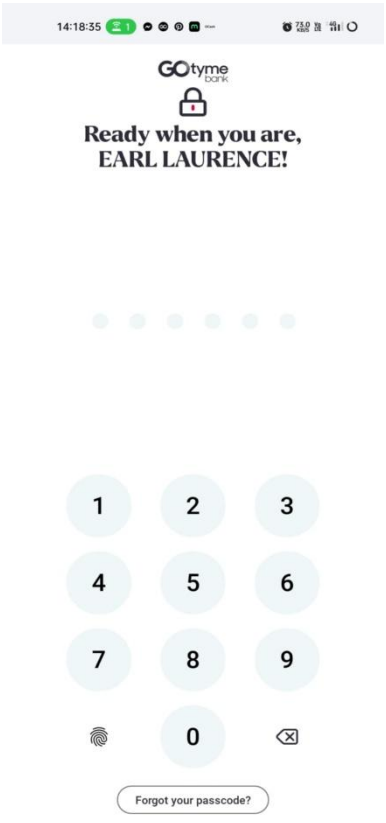
RECEIVED MONEY/TRANSACTION HISTORY





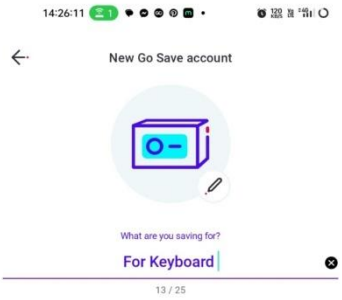
LOGOUT FUNCTIONALITY

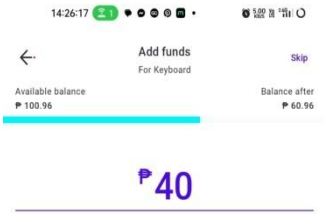




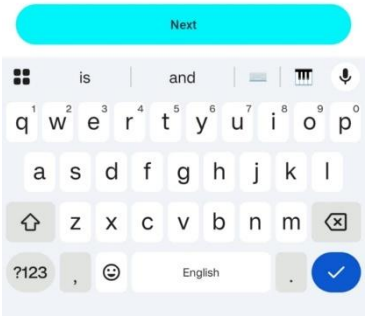
SAVING FUNCTIONALITY

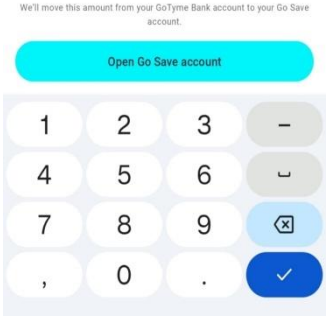


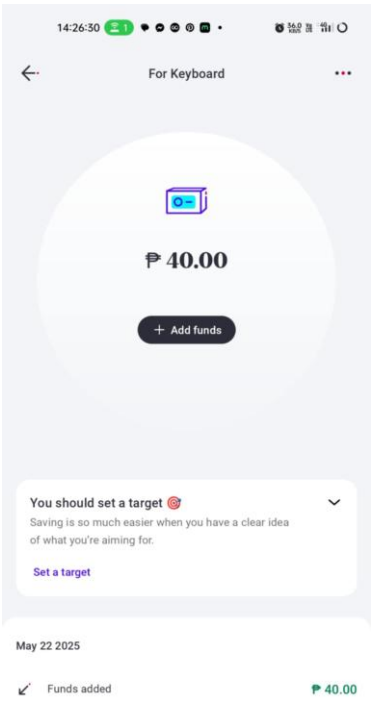


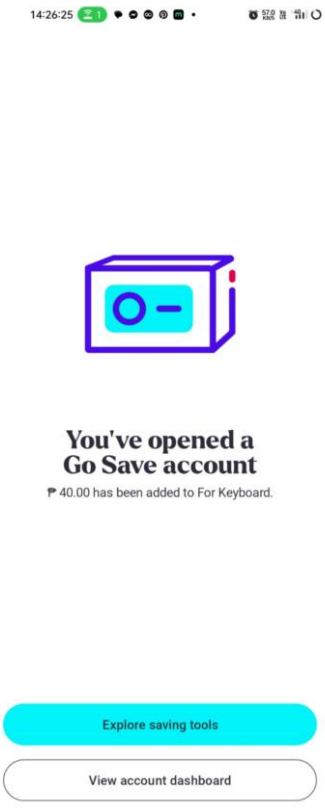






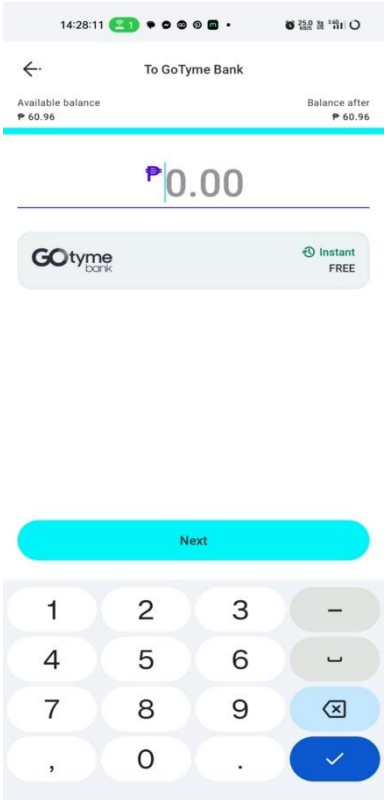




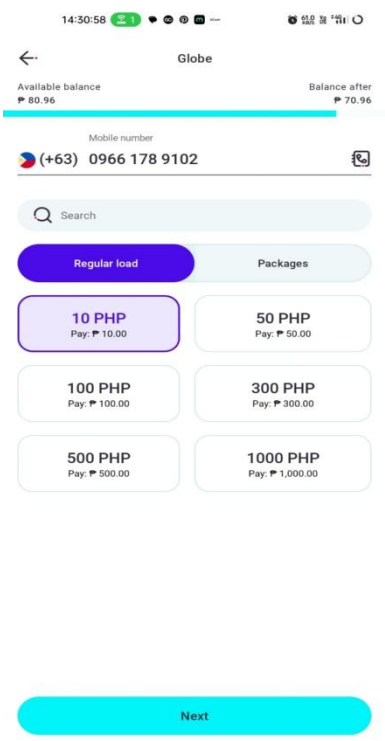


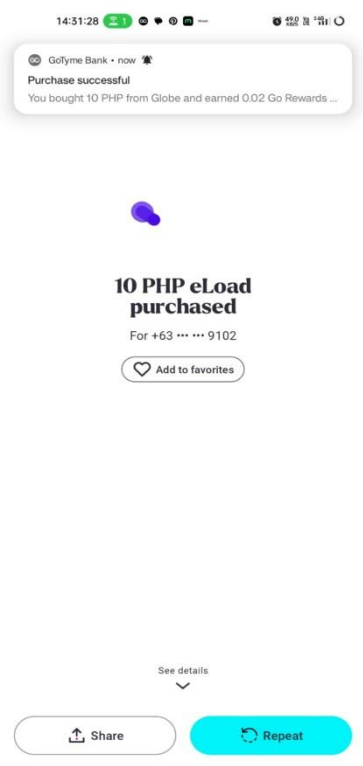
SCANNER

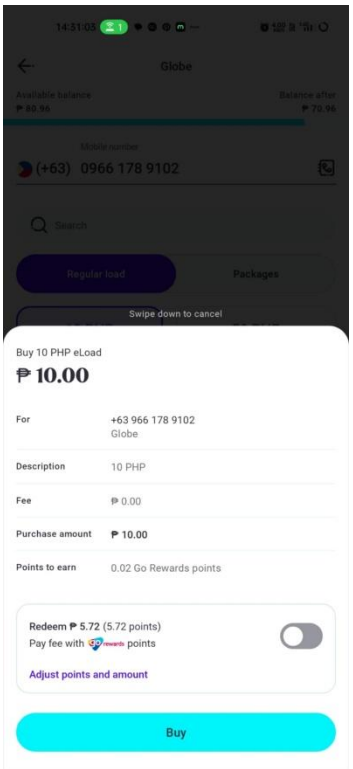




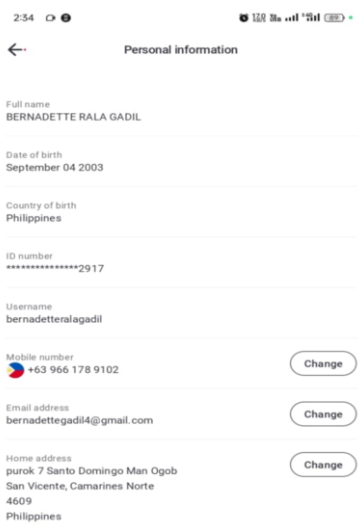
BUY LOAD FUNCTIONALITY



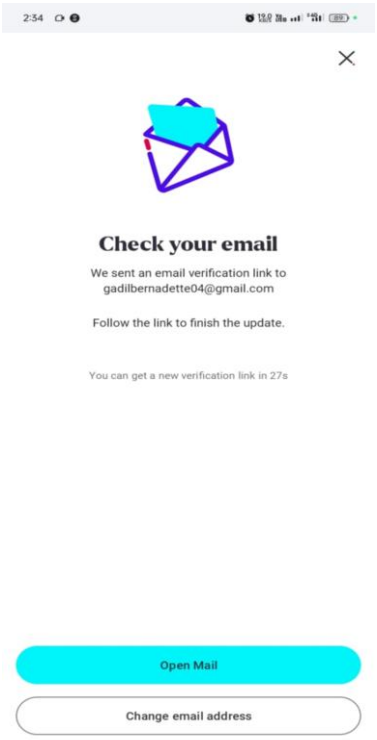




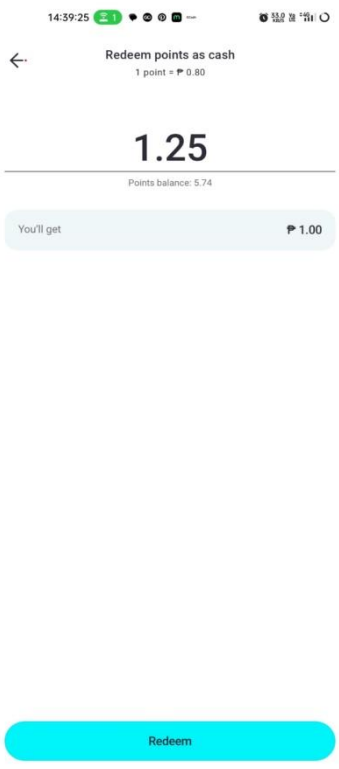
UPDATE INFORMATION

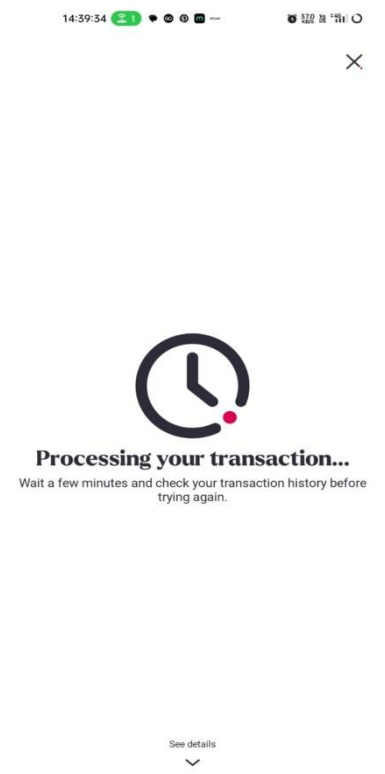


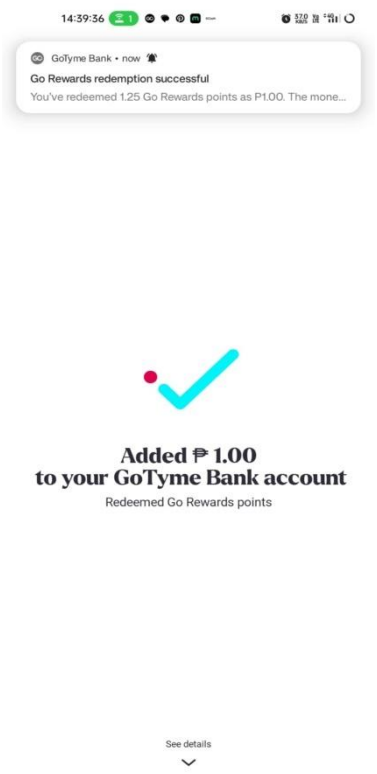




POINT CREDIT

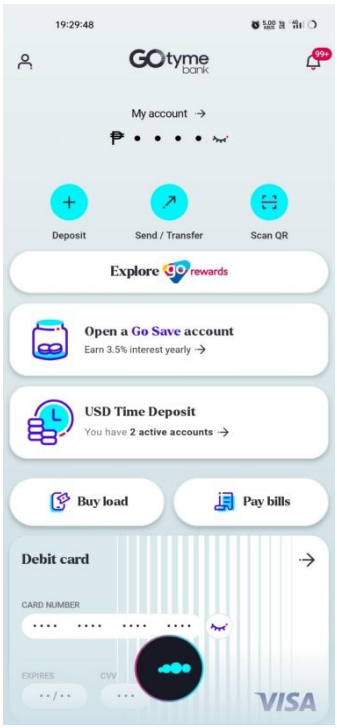


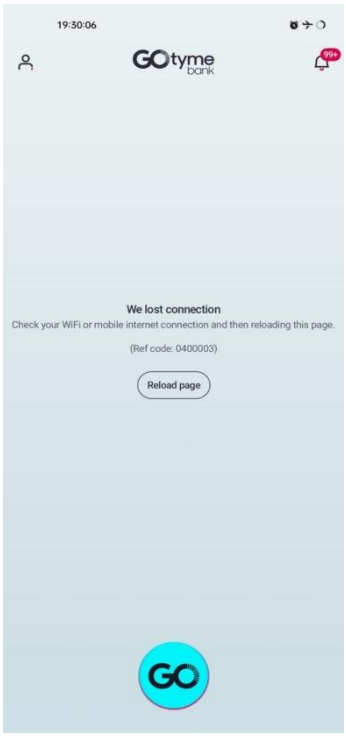




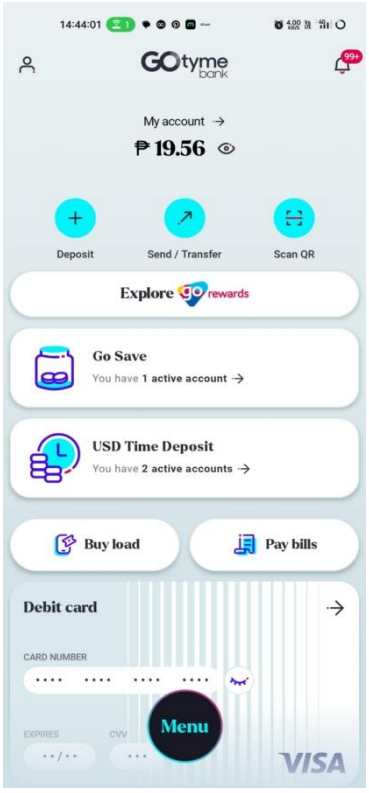
EXPLORATORY TESTING

AIRPLANE MODE

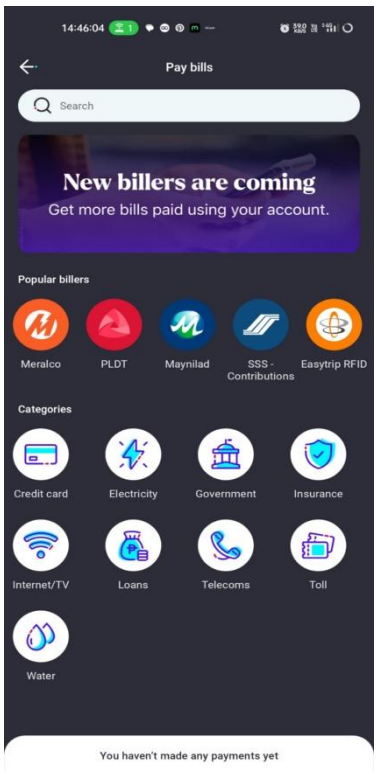


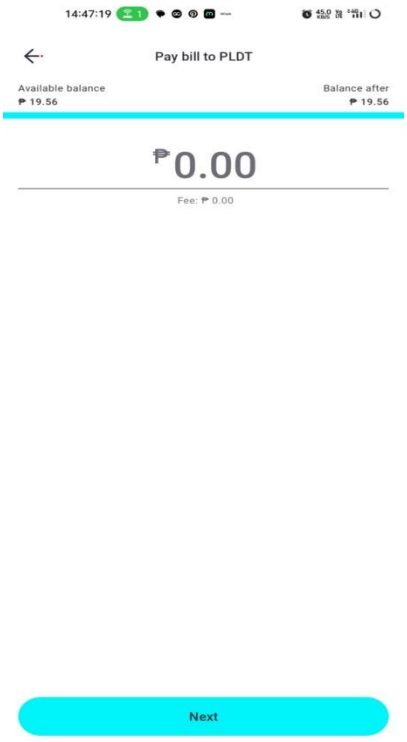


RAPIDLY TAPPING



REQUIRED VALUE

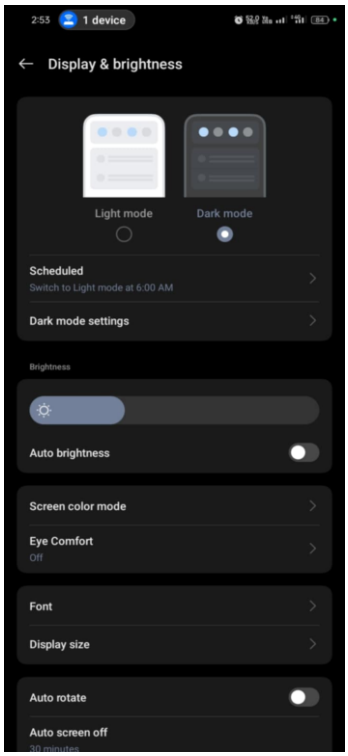




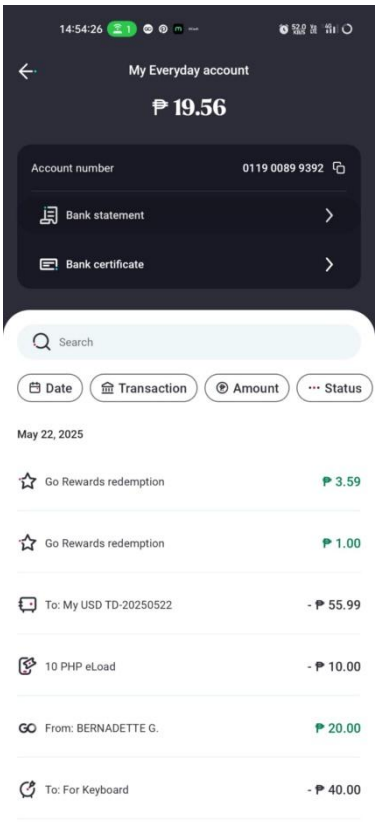


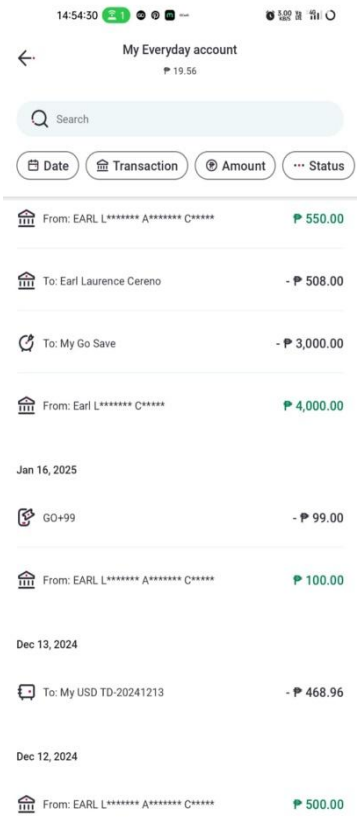
DARK MODE





TRANSACTION HISTORY



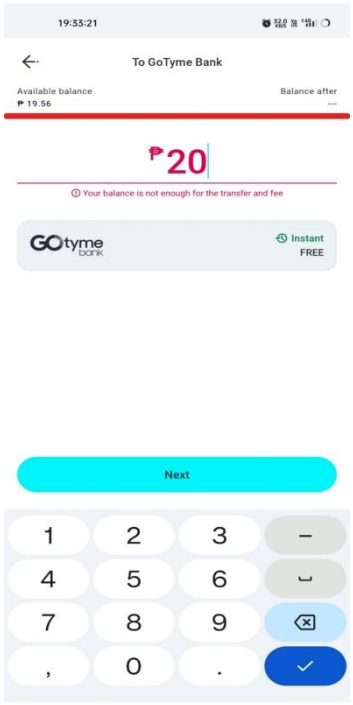


ATTEMPT TO LOGIN ACTIVE ACCOUNT TO ANOTHER DEVICE

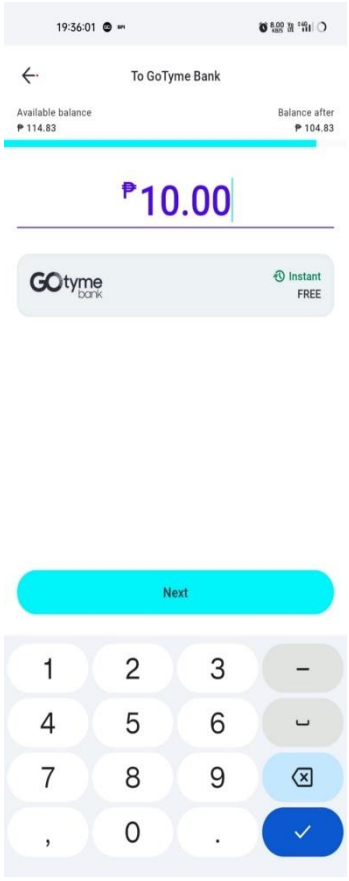




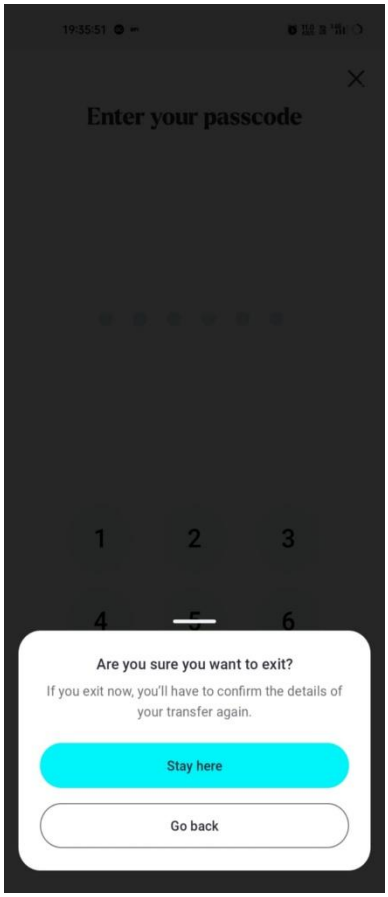
SEND MONEY HIGHER THAN ACC BALANCE



TAP BACK IN ONGOING TRANSACTION

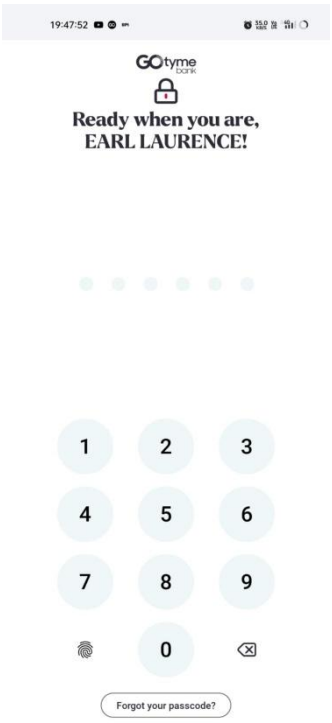


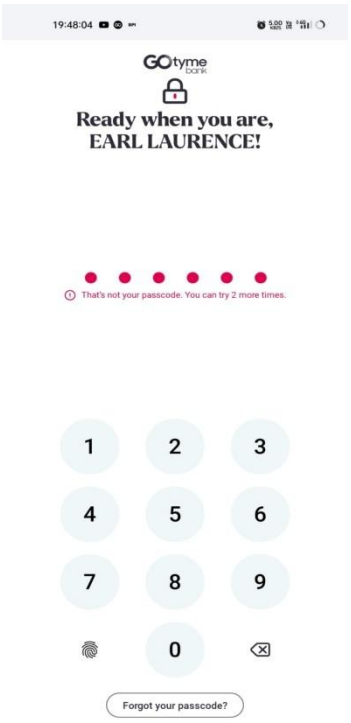


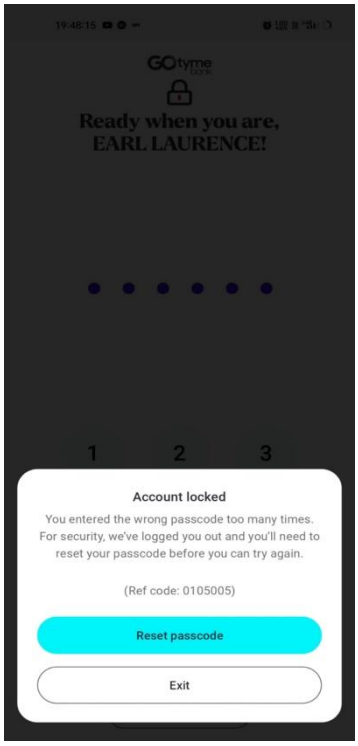


SECURITY TESTING

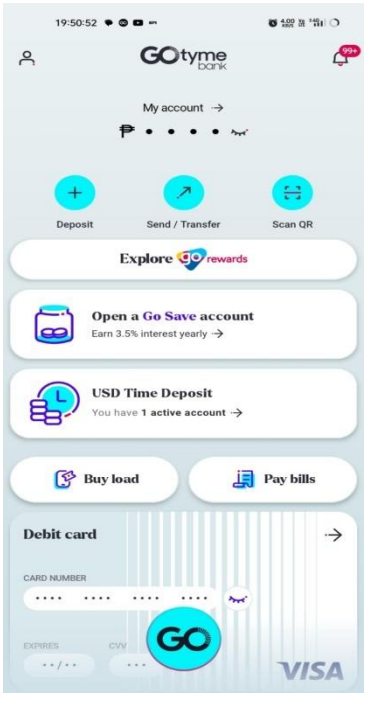
INCORRECT PASS

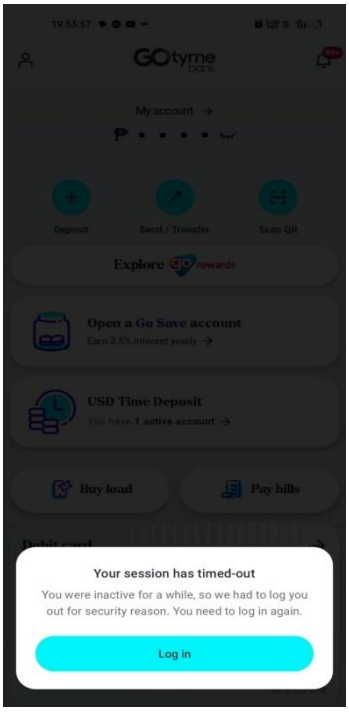




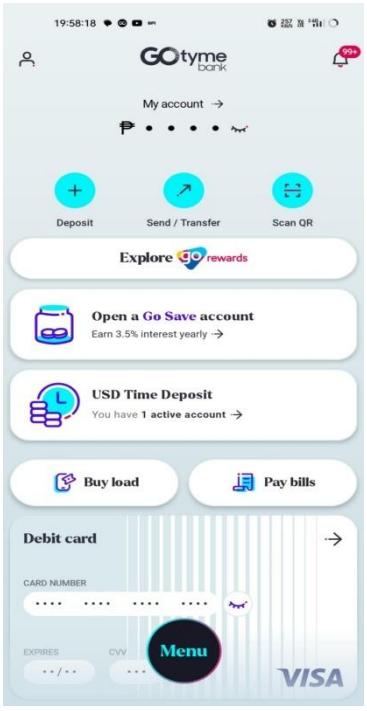


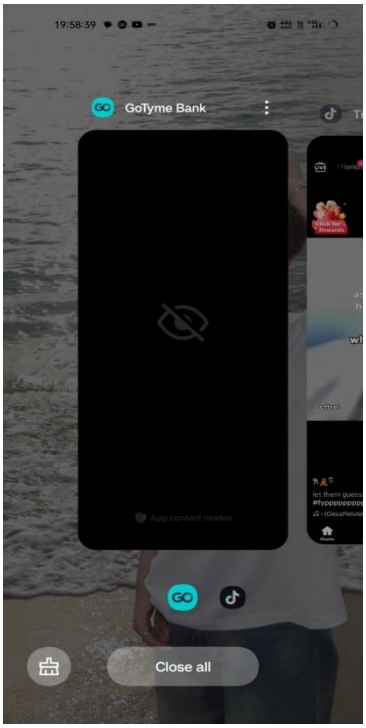
LEAVE THE APP AND RETURN IN 5MINS



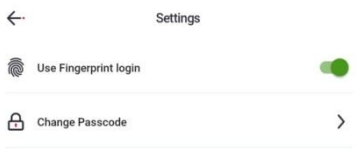


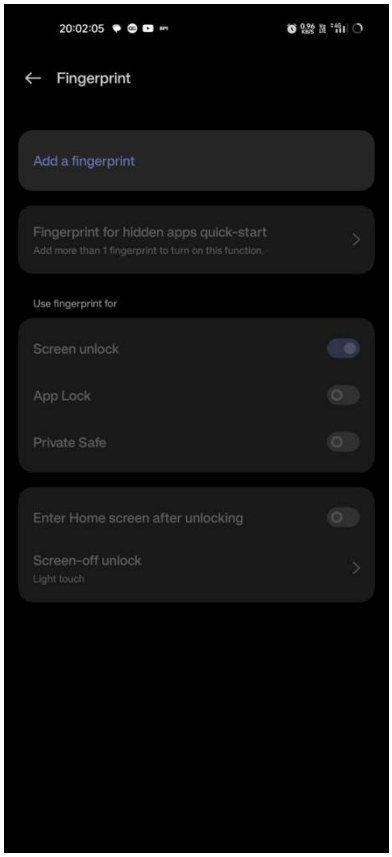
BYPASS LOGIN

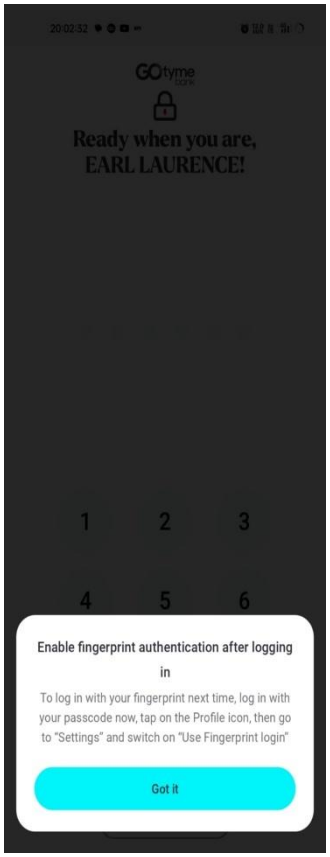




ENABLE BIOMETRIC LOGIN







SCREEN RECORD ON SENSITIVE INFO

